Voltage Level and Calculation

Welcome

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Voltage Level and Calculation

Voltage that is present on a line within limit is called acceptable voltage level.

Allowable Variation	Phase to neutral	Phase to phase
Maximum (105%)	241.5 Volt	418.3 Volt
Nominal (100%)	230.0 Volt	400.0 Volt
Minimum (96.5%)	221.9 Volt	386.0 Volt

Total system voltage permits variation
= Max^{m} service voltage near sub-station - Min^{m} service voltage at furthest end = 241.5 – 221.9 = 19.6 volt

Voltage Level and Calculation

Allowable Voltage drops (Max. load condition)

Max. Voltage 241.5 V on 230 V base

Sl.	Description	Max Voltage Drop (Volt)	% Voltage Drop(%)
1	Primary Line	6.9	3.0
2	Distribution Transformer	3.5	1.5
3	Secondary Main	4.6	2.0
4	Secondary Branch / Service	4.6	2.0
	Total	19.6	8.5

Voltage Level and Calculation

Effect of high voltage.

- Shorten life of electrical appliances;
- Damage of insulation of cable;
- Risk of life.

Effect of low voltage.

- Dim light;
- Improper operation of electrical appliances;
- Overheating of motor.

Benefits of acceptable level of voltage.

- Increased revenue;
- Greater consumer satisfaction;
- Higher efficiency of equipments;
- Lower investment per KVA.

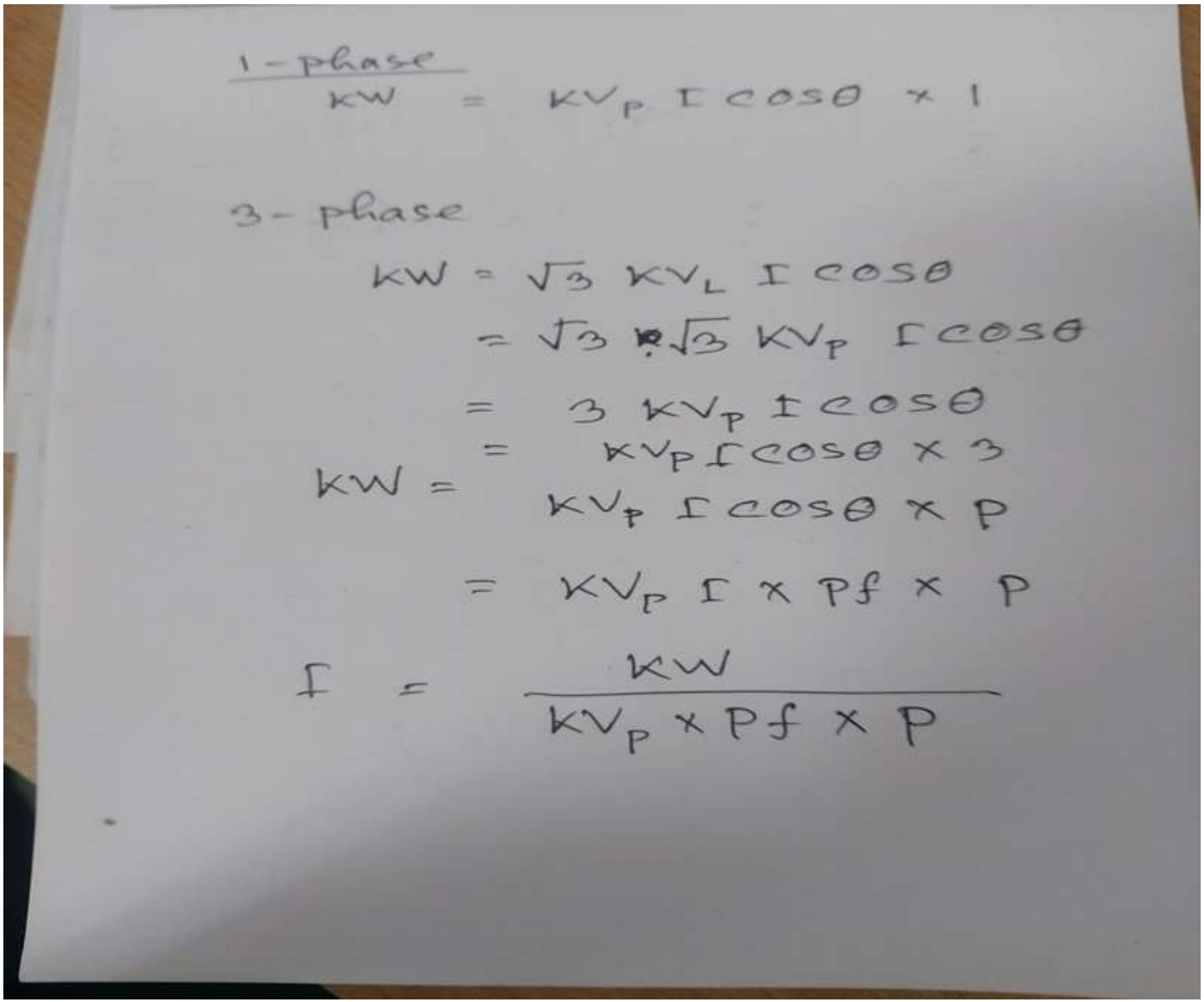
Voltage Level and Calculation

- Voltage Drop (VD)
-
- VD = I(R Cos A + X Sin A)
- Where I = Line current,
- A = Phase angle between voltage and current
- R = Resistance of line in ohms
- X = Reactance of line in ohms

- I = (KW) ÷ {(KV) × (Cos A) × (P)}
- Where KW = Circuit load in kilowatts,
- KV = System nom. phase to ground voltage in kilovolts
- P = Number of phases

- VD referred to 230v base = (Actual VD ÷ System Nom. Voltage) × 230

Voltage Level and Calculation



Voltage Level and Calculation

Voltage Drop (230 V base) = {(Actual Voltage Drop) × 230 }÷ System Nominal Voltage

If, Nominal System Voltage = 6.35/11 KV

Actual Voltage Drop = 360 V

Then VD(230 base) = (360 × 230) / 6350 = 13.04 V

Voltage Drop Calculation :

VD (230 V base) = (Actual VD × 230) ÷ Nominal System Voltage

$= (IZ \times 230) \div (KV \times 1000)$

$= [\{KW/(KV \times \cos\theta \times P)\} \times \{(R\cos\theta + X\sin\theta) \times S \times 230\}] \div [KV \times 1000]$

$= \{KW(R\cos\theta + X\sin\theta) \times S \times 230\} \div \{(KV)^2 \times \cos\theta \times P \times 1000\}$

$= (KW \times S \times VDF) \div 1000$

- Where r = Resistance in ohm per phase per km,
- x = Resistance in ohm per phase per km
- R = Reactance of line in ohms
- S = Distance in km

Where, $VDF = \{(R\cos\theta + X\sin\theta) \times 230\} \div \{(KV)^2 \times \cos\theta \times P\}$

Voltage Level and Calculation



Voltage Drop Calculation

Voltage drop = I (R cosθ+ X sinθ)
Where, I = Line current in amperes
θ = Phase angle between voltage and current
R = Resistance of line in ohms
X= Reactance of line in ohms

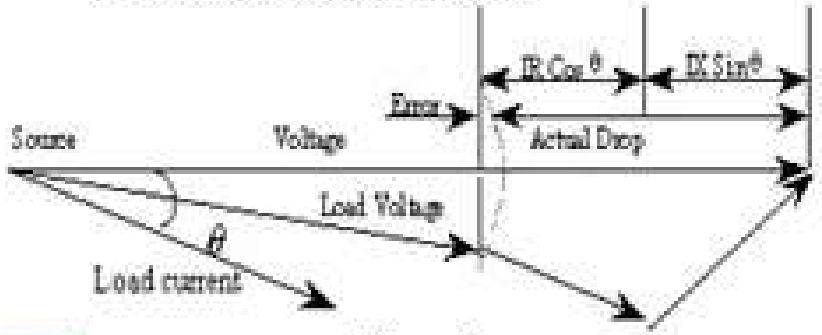


Figure 1.

Rural Electrification Board

Voltage Level and Calculation

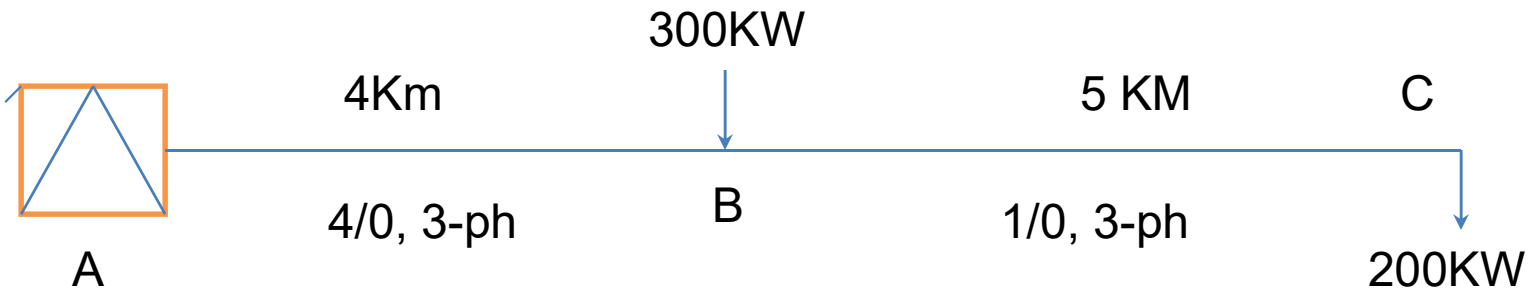
R & X is given bellow

Sl.	Name of Conductor	Three Phase		Single Phase	
		R	X	R	X
1	4/0	0.3522	0.4001	0.4737	0.6631
2	1/0	0.6709	0.4391	0.7982	0.7625
3	#3	1.2673	0.4417	1.3946	0.7650

Voltage Level and Calculation

For pf = 0.95, VDF Will be

Sl	Name of Conductor	VDF	
		Three Phase	Single Phase
1	4/0	0.9200	3.9500
2	1/0	1.5500	5.9800
3	#3	2.6900	9.3900



1. **What is the voltage drop at C?**

Answer:

VDF for 4/0, 3-ph = 0.92

VDF for 1/0, 3-ph = 1.55

$$VD_C = \frac{200 \times 5 \times 1.55}{1000} + \frac{500 \times 4 \times 0.92}{1000}$$

$$= (1.55+1.84)\ V$$

$$= 3.39\ \text{Volt.}$$

Voltage Level and Calculation

For design it is consider that Maximum Voltage Drop will be 3 % for any conductor and P.F is 0.95

For 4/0 ACSR, 3-Ph HT :

$VD = (KW \times S \times VDF) \div 1000$

Or, $3\%\ 6350 = (KW \times 1 \times 0.92) \div 1000$

(At 230V base)

Or, $\{(0.03 \times 6350 \times 230) \div 6350\} = \{(KW \times 1 \times 0.92) \div 1000\}$

Or, $KW\text{-}KM = (0.03 \times 6350 \times 1000 \times 230) \div (6350 \times 0.92)$

$= 7500$

So, $KVA\text{-}KM = 7500 \div 0.95$

$= 7894$

Hence , $MVA\text{-}KM = 7.894 \approx 8$

Voltage Level and Calculation

For #3 ACSR, 1-Ph LT :

$VD = IZ$

Or, $3\ \% \ 230 = (W \div V\cos\theta) \times (R\cos\theta + X\sin\theta)$

Or, $0.03 \times 230 = \{W \div (230 \times 0.95)\} \times \{(1.3946 \times 0.95) + (.765 \times .312)\}$

Or, $6.9 = (W \times 1.56) \div (230 \times 0.95)$

Or, $W\text{-}KM = (230 \times 0.95 \times 6.9) \div 1.56$

$= 966.44$

Or, $KW\text{-}KM = 0.966$

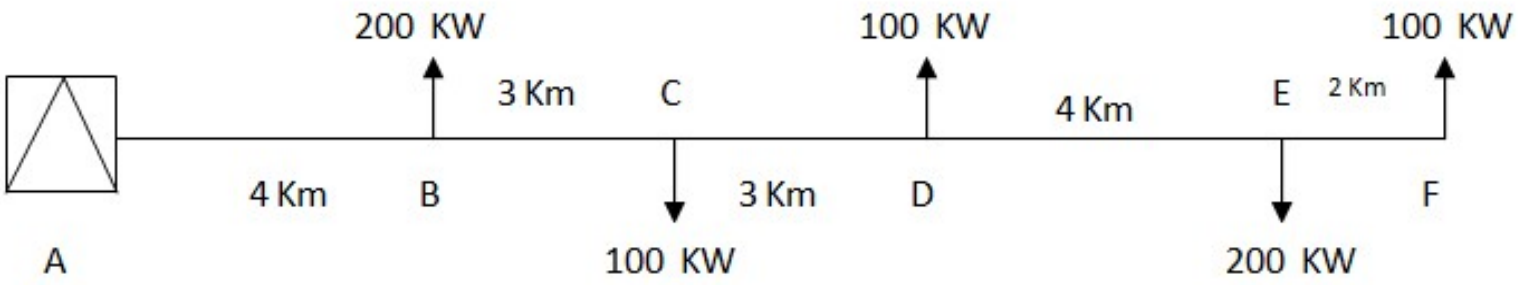
Or, $KVA\text{-}KM = (0.966 \div 0.95)$

$= 1.01 \approx 1$

Voltage Level and Calculation

MVA/KVA-KM Table

Sl No.	Size of Conductor	MVA/KVA-Km
1	4/0 ACSR – 3-ϕ HT	8 MVA - Km
2	1/0 ACSR – 3-ϕ HT	5 MVA - Km
3	# 3 ACSR – 3-ϕ HT	2.5 MVA - Km
4	# 3 ACSR – 1-ϕ HT	800 KVA - Km
5	# 3 ACSR – 3-ϕ LT	2.5 KVA - Km
6	# 3 ACSR – 1-ϕ LT	1 KVA - Km
7	# 3 Quadruplex – LT	2.5 KVA - Km
8	# 3 Duplex – LT	1 KVA - Km
9	1/0 Quadruplex – LT	5 KVA – Km
10	477 MCM ACSR – 3-ϕ HT	16 MVA - Km



KW-KM = (100 x 2) + (300 x 4) + (400 x 3) + (500 x 3) + (700 x 4)
= 200 + 1200 + 1200 + 1500 + 2800
= 6900

KVA-KM= 6900/0.95 = 7263

MVA-KM = 7.26

Therefore, **Conductor Size = 4/0 ACSR**

Voltage Level and Calculation

Questions????